Joint Neural Demodulation and Advanced FEC (LDPC/Polar) over OFDM

Overview and Motivation

Modern 5G/6G systems rely heavily on Orthogonal Frequency Division Multiplexing (OFDM) combined with powerful Forward Error Correction (FEC) codes, specifically Low-Density Parity-Check (LDPC) and Polar codes. To push a communication link close to the theoretical Shannon capacity limit, the demodulator cannot just output binary 1s and 0s. It must provide the downstream FEC decoder with highly accurate soft confidence scores, known as Log-Likelihood Ratios (LLRs).

The fundamental problem is that traditional mathematical demappers calculate LLRs based on the assumption of a perfect Gaussian noise environment. In reality, signals transmitted over OFDM suffer from frequency-selective multipath fading, Doppler shifts, and hardware interference. Consequently, traditional demappers generate inaccurate LLRs—they are overly confident in wrong bits or under-confident in right bits. When these "bad" LLRs are fed into a Belief Propagation (BP) decoder, the error correction fails prematurely.

This project bridges the gap between deep learning and classical error correction. The intern will train a Convolutional Neural Network (CNN) to "read" the distorted 2D time-frequency OFDM resource grid. Because the neural network can learn the *true*, complex noise distribution of the channel, it will generate highly calibrated, optimized LLRs tailored specifically to maximize the success rate of the subsequent LDPC decoding process.

Technical Objectives

- **OFDM-FEC Pipeline Integration:** Deploy a complete, standards-compliant OFDM transmission chain that flawlessly integrates Sionna's 5G NR LDPC encoders and decoders.
- **Neural Demapper Architecture:** Design a CNN capable of processing the multi-dimensional structure of OFDM resource grids (treating subcarriers and OFDM symbols as spatial dimensions).
- **LLR Calibration:** Ensure the output of the neural demapper is properly scaled and calibrated as probabilistic LLRs, matching the mathematical input requirements of Sionna's differentiable BP decoder.
- **Joint Differentiable Training:** Perform end-to-end training utilizing backpropagation *through* the FEC decoder (if differentiable) or using proxy loss functions, ensuring the neural demapper is optimized specifically for post-FEC BER minimization, not just raw symbol error rate.

Milestones

- **Task 1 (OFDM & FEC Framework):** Build the full OFDM transmission chain. Integrate Sionna's LDPC encoder (`sionna.fec.ldpc.LDPC5GEncoder`), map the encoded bits to QAM, and map those to the OFDM resource grids.
- **Task 2 (Channel Simulation):** Pass the generated signal through a standard 3GPP frequency-selective multipath fading channel (e.g., `sionna.channel.tr38901.TDL`). Establish the baseline BER using classical demappers and LDPC decoders.
- **Task 3 (CNN Demapper Construction):** Replace the standard demapper with a Convolutional Neural Network. Train the CNN to ingest the distorted OFDM grid and output soft LLRs.
- **Task 4 (Joint Optimization & Delivery):** Feed the CNN-generated LLRs directly into Sionna's differentiable LDPC decoder (`sionna.fec.ldpc.LDPC5GDecoder`). Fine-tune the CNN weights to minimize the ultimate post-FEC BER. Deliver the architecture and performance reports.